DGMS 2019 • SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Question 74: Irrigation Water Quality: Sodium Adsorption Ratio (SAR) & Soil Dispersion

Base Exchange Thermodynamics, Clay Deflocculation Physics, and Permeability Degradation

OFFICIAL EXAMINATION QUESTION

Consider the following statements regarding irrigation water:

Statement 1: Sodium concentration is important for classifying irrigation water.

Statement 2: Sodium reacts with soil to reduce its permeability and hence affects the agricultural activity.

Which one of the following is correct in respect of the above statements?

(a) Both Statement 1 and Statement 2 are true and Statement 2 is the correct explanation of Statement 1 [CORRECT]

(b) Both are true but Statement 2 is not the correct explanation of Statement 1

(c) Statement 1 is true but Statement 2 is false

(d) Statement 1 is false but Statement 2 is true

VERIFIED ANSWER KEY

Correct Option: (a) Both Statement 1 and Statement 2 are true and Statement 2 is the correct explanation of Statement 1 — In agricultural hydrogeology, sodium concentration is paramount because excessive $\mathrm{Na^+}$ replaces divalent calcium ($\mathrm{Ca^{2+}}$) and magnesium ($\mathrm{Mg^{2+}}$) on clay mineral exchange sites. This base-exchange reaction expands the diffuse double layer, causing clay particles to disperse (deflocculate). The dispersed clay platelets clog soil pore throats, drastically reducing soil hydraulic conductivity and infiltration, creating waterlogging, and

impeding root respiration. Thus, Statement 2 is the direct causal explanation of Statement 1.

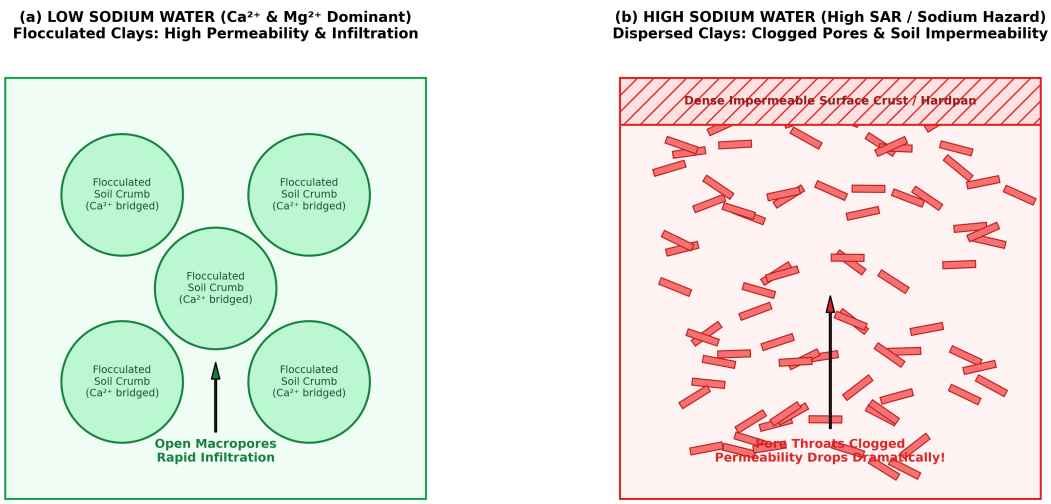


Figure 74.1 | Soil Physical Degradation under High Sodium Irrigation Water: (a) **Low Sodium / High Calcium Water:** Divalent $\mathrm{Ca^{2+}}$ cations bind negatively charged clay platelets into stable, flocculated soil crumbs, maintaining open macropores for rapid drainage and root aeration. (b) **High Sodium Hazard (High SAR):** Monovalent $\mathrm{Na^+}$ displaces $\mathrm{Ca^{2+}}$, swelling and dispersing individual clay platelets that clog pore throats, forming a dense, impermeable surface crust that destroys agricultural productivity.

1. Soil Chemistry & The Mechanism of Sodium Hazard

Soils contain clay minerals (smectite, illite, kaolinite) and organic matter that carry permanent negative surface charges. In fertile agricultural soils, these negative charges are predominantly balanced by **divalent cations ($\mathrm{Ca^{2+}}$ and $\mathrm{Mg^{2+}}$)**:

- Divalent cations possess a small hydrated ionic radius and high charge density, tightly compressing the electrical diffuse double layer around clay particles.
- This allows van der Waals attractive forces to pull clay platelets together into stable aggregated clusters or "crumbs" (**flocculation**). Flocculated soils exhibit high porosity, excellent crumb structure, rapid infiltration, and high permeability.

When irrigation water with high sodium concentration is applied, a cation exchange reaction occurs:

$$\mathrm{Ca\text{-}Clay + 2Na^+ \rightleftharpoons 2Na\text{-}Clay + Ca^{2+}}$$

- Monovalent $\mathrm{Na^+}$ ions have a larger hydrated radius and lower charge density. As sodium occupies more than 15% of the cation exchange capacity (Exchangeable Sodium Percentage, $\text{ESP} > 15\%$), the diffuse double layer expands dramatically.

- Electrostatic repulsive forces overcome van der Waals attraction: clay clusters break apart into microscopic platelets in a destructive process called **deflocculation (dispersion)**.
- Dispersed clay platelets wash down into pore throats, completely blocking them and reducing soil hydraulic conductivity by **two to three orders of magnitude**! Upon drying, the surface bakes into an impenetrable, hard crust.

2. Quantitative Indices of Sodium Hazard in Irrigation Water

Agricultural hydrologists classify irrigation water using standardized empirical parameters:

1. Sodium Adsorption Ratio (SAR) — US Salinity Laboratory (1954)

The standard index quantifying the relative activity of sodium ions in exchange reactions with soil:

$$\text{SAR} = \frac{[\mathrm{Na^+}]}{\sqrt{\frac{[\mathrm{Ca^{2+}}] + [\mathrm{Mg^{2+}}]}{2}}}$$

where all ionic concentrations are expressed in milliequivalents per liter (meq/L).

SAR Value Range	Sodium Hazard Class	Suitability & Agricultural Impact
0 – 10	**S1 (Low Sodium)**	Safe for almost all crops and all soils; minimal hazard of clay dispersion.
10 – 18	**S2 (Medium Sodium)**	Appreciable hazard in fine-textured soils; usable on coarse sandy soils with high leaching.
18 – 26	**S3 (High Sodium)**	Harmful to most soils; produces severe dispersion and drainage problems; requires gypsum ($\mathrm{CaSO_4}$) amendments.
> 26	**S4 (Very High Sodium)**	Unsuitable for irrigation; causes complete soil structural breakdown and alkaline toxicity.

2. Percent Sodium (%Na)

$$\% \mathrm{Na} = \frac{[\mathrm{Na^+}] + [\mathrm{K^+}]}{[\mathrm{Ca^{2+}}] + [\mathrm{Mg^{2+}}] + [\mathrm{Na^+}] + [\mathrm{K^+}]} \times 100$$

Water with $\% \mathrm{Na} > 60\%$ is considered hazardous for agriculture due to rapid alkali development.

3. Logical Relationship: Statement 1 & Statement 2

- **Statement 1:** True. Sodium concentration is a vital criterion in all global irrigation classification schemes (USSL diagram, Wilcox diagram).
- **Statement 2:** True. Sodium directly induces clay dispersion, destroying macropores and slashing soil permeability.
- **Causal Link:** Statement 2 provides the exact physical and chemical rationale for Statement 1. We classify irrigation water based on sodium *because* sodium degrades soil permeability and harms crops.

4. Authoritative Textbook References

- **David Keith Todd & Larry W. Mays (2005)** *Groundwater Hydrology*, 3rd Ed., John Wiley & Sons, Chapter 8 ("Quality of Groundwater: Suitability for Irrigation / SAR"), pp. 340–355.
- **L. A. Richards (ed.) (1954)** *Diagnosis and Improvement of Saline and Alkali Soils*, USDA Agriculture Handbook No. 60 (US Salinity Laboratory Staff), Washington D.C.
- **C. W. Fetter (2001)** *Applied Hydrogeology*, 4th Ed., Prentice Hall, Chapter 9 ("Groundwater Chemistry: Agricultural Water Standards").
- **H. M. Raghunath (2007)** *Ground Water*, 3rd Ed., New Age International Publishers, Chapter 7 ("Irrigation Water Quality Criteria").